# 8086 MEMORY BANKS

→ Concept used to make data processing efficient
→ A way of organizing memory so that up can efficiently handle its 16 bits data operations

8086 provides 16 bits data bus – so it is **capable** of transferring 16 bits = 2 bytes in 1 cycle but each memory location is only of 1 byte = 8 bit

↳ So, we need to use 2 cycles to access 2 bytes (8 bits each) from 2 diff" memory locations.

↳ Soln: **MEMORY BANK**

↳ goal: to access two consecutive mem. locs in 1 cycle
↳ to transfer 16 bits in 1 cycle

Each addr. can store 8 bits = 1 byte data

| Addr. | Data |
|---|---|
| 00000 | 34 h |
| 00001 | 12h |
| 00002 | |
| 00003 | |

AX = | AH = 12 | AL = 34 |

T1: Addr. Bus → How can we send addr. in T1 cycle so that 16 bit data can be fetched in 1 cycle?
T2: Bus cycle
T3: Data transfer
T4: Sys. reset

↓
MEM. BANK

Addr. → A0-15 & A16-19

| $A_{19}$ . . . $A_2$ | $A_1$ | $A_0$ | Address | |
|---|---|---|---|---|
| 0 . . . 0 | 0 | 0 | 00000 | A1-19 equal, A0 = 0 (even) = 1 (odd) |
| 0 . . . 0 | 0 | 1 | 00001 | |
| 0 . . . 0 | 1 | 0 | 00002 | A1-B equal, A0 = 0 (even) = 1 (odd) |
| 0 . . . 0 | 1 | 1 | 00003 | |

**point 1** A0 = 0 (even), A0 = 1 (odd)

① RD/WR from both odd & even banks
② " " odd bank
③ " " even bank
④ don't fetch

4 states → cannot be represented by 1 bit
↳ needs 2 bits = $2^2 = 4$ states

$\overline{BHE}$ + A0

**point 2**

| $\overline{BHE}$ | A0 | |
|---|---|---|
| 0 | 0 | → both banks (16 bit) |
| 0 | 1 | → odd bank (D8-15) |
| 1 | 0 | → even bank (D7-0) |
| 1 | 1 | → none/ IDLE |

**point 3** ONLY **aligned data** can be addressed in 1 bus cyc.

00000 → 00001 } aligned
00002 → 00003

00001 } unaligned
00002

MEMORY (1 MB)

512KB even bank (lower bank): 00000, 00002, 00004, ⋮, FFFFE
512KB odd bank (higher bank): 00001, 00003, 00005, ⋮, FFFFF

$\overline{BHE}$ + A0

MOV AL, [00002h] → 1 BC = 10
MOV AL, [00003h] → 1 BC = 01
MOV AX, [0002h] → 1 BC = 00
MOV AX, [0001h] → not 1 BC → 2 BC

addr. = odd no + 16 bit data transfer
req. 2 Bus cycle

| | $\overline{BHE}$ | A0 |
|---|---|---|
| ↳ 1st BC: | 0 | 1 |
| 2nd BC: | 1 | 0 |